If - elif - else statement

In [1]:
```
#W. A P. which takes one number from 0 to 9 from the user and prints it in the word
#it should print that number is outside of the range and program should exit.
# For exapmple:-
# input = 1
# output = one

num = int(input('Enter a number: '))
if num == 0:
  print('Zero')
elif num ==1:
  print('One')
elif num ==2:
  print('Two')
elif num ==4:
  print('Four')
elif num ==5:
  print('Five')
elif num ==6:
  print('Six')
elif num ==7:
  print('Seven')
elif num ==8:
  print('Eight')
elif num ==9:
  print('Nine')
else:
  print('Number is outside the range!!!')


print("\n ---------- \n")

# another method

number = int(input("Enter a number between 0 and 9: "))
if number <0 or number > 9:
    print("The number is outside of the range")
else:
    words = ["zero", "one", "two", "three", "four", "five", "six", "seven", "eight"
    print(words[number])
```

```
Enter a number: 5
Five

 ----------
```

In [1]:
```
#W. A P. to implement calculator but the operation to be done and two numbers will
#Operation console should show below:-
#     Please select any one operation from below:-
#         * To add enter 1
#         * to subtract enter 2
```

```python
#         * To multiply enter 3
#         * To divide enter 4
#         * To divide and find quotient enter 5
#         * To divide and find remainder enter 6
#         * To divide and find num1 to the power of num2 enter 7
#         * To Come out of the program enter 8
import sys
num1 = int(input('Enter 1st number: '))
num2 = int(input('Enter 2nd number: '))

op = int(input('''Please select any one operation from below:-\n
  \tTo add enter 1\n
  \tTo subtract enter 2\n
  \tTo multiply enter 3\n
  \tTo divide enter 4\n
  \tTo divide and find quotient enter 5\n
  \tTo divide and find remainder enter 6\n
  \tTo divide and find num1 to the power of num2 enter 7\n
  \tTo exit Enter 8\n
  '''))
if op == 1:
    print(num1+num2)
elif op == 2:
    print(num1-num2)
elif op == 3:
    print(num1*num2)
elif op == 4:
    print(num1/num2)
elif op == 5:
    print(num1//num2)
elif op == 6:
    print(num1%num2)
elif op == 7:
    print(num1**num2)
elif op == 8:
    sys.exit(0)
else:
    print('wrong input\nTry Again\n\n')
```

```
Enter 1st number: 5
Enter 2nd number: 6
Please select any one operation from below:-

        To add enter 1

        To subtract enter 2

        To multiply enter 3

        To divide enter 4

        To divide and find quotient enter 5

        To divide and find remainder enter 6

        To divide and find num1 to the power of num2 enter 7

        To exit Enter 8

  1
11
```

In [6]:

```python
# Another method

operation = input('''
Please type in the math operation you would like to complete:
+ for addition
- for subtraction
* for multiplication
/ for division
''')

number_1 = int(input('Enter your first number: '))
number_2 = int(input('Enter your second number: '))

if operation == '+':
    print('{} + {} = '.format(number_1, number_2))
    print(number_1 + number_2)

elif operation == '-':
    print('{} - {} = '.format(number_1, number_2))
    print(number_1 - number_2)

elif operation == '*':
    print('{} * {} = '.format(number_1, number_2))
    print(number_1 * number_2)

elif operation == '/':
    print('{} / {} = '.format(number_1, number_2))
    print(number_1 / number_2)

else:
    print('You have not typed a valid operator, please run the program again.')
```

```
Please type in the math operation you would like to complete:
+ for addition
- for subtraction
* for multiplication
/ for division
*
Enter your first number: 6
Enter your second number: 4
6 * 4 =
24
```

In [7]:

```python
# Using DEF function

def calculate():
    operation = input('''
Please type in the math operation you would like to complete:
+ for addition
- for subtraction
* for multiplication
/ for division
''')

    number_1 = int(input('Please enter the first number: '))
    number_2 = int(input('Please enter the second number: '))

    if operation == '+':
        print('{} + {} = '.format(number_1, number_2))
        print(number_1 + number_2)

    elif operation == '-':
        print('{} - {} = '.format(number_1, number_2))
        print(number_1 - number_2)

    elif operation == '*':
        print('{} * {} = '.format(number_1, number_2))
        print(number_1 * number_2)

    elif operation == '/':
        print('{} / {} = '.format(number_1, number_2))
        print(number_1 / number_2)

    else:
        print('You have not typed a valid operator, please run the program again.')

    # Add again() function to calculate() function
    again()

def again():
    calc_again = input('''
Do you want to calculate again?
Please type Y for YES or N for NO.
''')

    if calc_again.upper() == 'Y':
        calculate()
    elif calc_again.upper() == 'N':
```

```
        print('See you later.')
    else:
        again()

calculate()
```

```
Please type in the math operation you would like to complete:
+ for addition
- for subtraction
* for multiplication
/ for division
+
Please enter the first number: 1
Please enter the second number: 2
1 + 2 =
3

Do you want to calculate again?
Please type Y for YES or N for NO.
y

Please type in the math operation you would like to complete:
+ for addition
- for subtraction
* for multiplication
/ for division
-2
Please enter the first number: 2
Please enter the second number: 3
You have not typed a valid operator, please run the program again.

Do you want to calculate again?
Please type Y for YES or N for NO.
n
See you later.
```

In [14]:

```
#W A P to check whether a year entered by user is an leap year or not?
#Check with below input:-
    #leap year:- 2012, 1968, 2004, 1200, 1600,2400
    #Non-lear year:- 1971, 2006, 1700,1800,1900

# Using IF and Else

year=int(input("Enter year to be checked:"))
if(year%4==0 and year%100!=0 or year%400==0):
    print("The year is a leap year")
else:
    print("The year isn't a leap year")
```

```
Enter year to be checked:1971
The year isn't a leap year
```

In [3]:

```
# Python program to check if year is a leap year or not

# To get year (integer input) from the user
year = int(input("Enter a year: "))
```

```
# divided by 100 means century year (ending with 00)
# century year divided by 400 is leap year
if (year % 400 == 0) and (year % 100 == 0):
    print("{0} is a leap year".format(year))

# not divided by 100 means not a century year
# year divided by 4 is a leap year
elif (year % 4 ==0) and (year % 100 != 0):
    print("{0} is a leap year".format(year))

# if not divided by both 400 (century year) and 4 (not century year)
# year is not leap year
else:
    print("{0} is not a leap year".format(year))
```

```
Enter a year: 1800
1800 is not a leap year
```

In [21]:

```
#W A P which takes one number from the user and checks whether it is an even or odd
#even number else prints that number is odd number.

# Using modolus method
num = int(input("Enter any number to test whether it is odd or even: "))
if (num % 2) == 0:
              print ("The number is even")
else:
              print ("The provided number is odd")
```

```
Enter any number to test whether it is odd or even: 5
The provided number is odd
```

In [24]:

```
#W A P which takes two numbers from the user and prints below output:-
#     1. num1 is greater than num2 if num1 is greater than num2
#     2. num1 is smaller than num2 if num1 is smaller than num2
#     3. num1 is equal to num2 if num1 and num2 are equal

#Note:- 1. Do this problem using if - else
#       2. Do this using ternary operator
num_1 = int(input("Please enter the first number :"))
num_2 = int(input("Please enter the first number :"))

# condition 1 using if - else method
if num_1 > num_2:
    print("\n The first number is greater that the second \n")
elif num_1 < num_2:
    print("\n The second number is greater that the first \n")
else: print(" Both the numbers are equal")
```

```
Please enter the first number :2
Please enter the first number :5

 The second number is greater that the first

Please enter the first number :6
Please enter the second number :4
Please enter the third number :6
Largest number among 6 , 4 and 6 is 6
```

In [5]:
```python
# Using Ternary operator

n1 = int(input("Please enter the first number :"))
n2 = int(input("Please enter the second number :"))

mx = n1 if n1 > n2 else n2

print(mx)
```

```
Please enter the first number :44
Please enter the second number :45
45
```

In [1]:
```python
#W A P which takes three numbers from the user and prints below output:-
#     1. num1 is greater than num2 and num3 if num1 is greater than num2 and num3
#     2. num2 is greater than num1 and num3 if num2 is greater than num1 and num3
#     3. num3 is greater than num1 and num2 if num3 is greater than num1 and num2

#Note:- 1. Do this problem using if - elif - else
#       2. Do this using ternary operator

# a = a if a>b else b
#expr if cond1 else expr2 if cond2 else expr3


n1 = int(input("Please enter the first number :"))
n2 = int(input("Please enter the second number :"))
n3 = int(input("Please enter the third number :"))

mx = (n1 if (n1 > n2 and n1 > n3) else
     (n2 if (n2 > n1 and n2 > n3) else n3))

print(mx)
```

```
Please enter the first number :56
Please enter the second number :45
Please enter the third number :68
68
```

Loops - for loop, while loop

In [10]:
```python
#Write a Python program to find the length of the my_str using loop:-

#Input:- 'Write a Python program to find the length of the my_str'
```

```
#Output:- 55

my_str = "Write a Python program to find the length of the my_str"

counter = 0
for i in my_str:
      counter = counter+1
print("Length of the input string is:", counter)
```

```
Length of the input string is: 55
```

In [14]:
```
#Write a Python program to find the total number of times letter 'p' is appeared in

#Input:- 'peter piper picked a peck of pickled peppers.\n'
#Output:- 9

my_str = 'peter piper picked a peck of pickled peppers.\n'
count = 0
for i in my_str:
    if i == 'p':
        count = count + 1
print ("Count of p in my_str is : "+  str(count))
```

```
Count of p in my_str is : 9
```

In [6]:
```
#Write a Python Program, to print all the indexes of all occurences of letter 'p' a

#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:-
# 0
# 6
# 8
# 12
# 21
# 29
# 37
# 39
# 40

my_str = 'peter piper picked a peck of pickled peppers.'
s='p'
for i in range(len(my_str)):
    if (my_str[i] == s):
        print(i)
```

```
0
6
8
12
21
29
37
39
40
```

In [8]:
```
#Write a python program to find below output using loop:-
```

```python
#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:- ['peter', 'piper', 'picked', 'a', 'peck', 'of', 'pickled', 'peppers']

# Using .split()
my_str = 'peter piper picked a peck of pickled peppers.'
print(my_str.split())


print('--------\n')

myStr = 'peter piper picked a peck of pickled peppers.'
output = []

start = 0
for i in range(0, len(myStr)):
    if(myStr[i]==" ") or myStr[i]==".":
        output.append(myStr[start:i])
        start = i+1
print(output)


#2nd method

new_str = ''
lst = []
for i in my_str:
    if i==' 'or i=='.':
        lst+=[new_str]
        new_str=''
    else:
        new_str+=i
print(lst)
```

```
['peter', 'piper', 'picked', 'a', 'peck', 'of', 'pickled', 'peppers.']
--------

['peter', 'piper', 'picked', 'a', 'peck', 'of', 'pickled', 'peppers']
```

In [14]:

```python
#Write a python program to find below output using loop:-

#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:- 'peppers pickled of peck a picked piper peter'


myStr = 'peter piper picked a peck of pickled peppers.'
output =''
start = 0
for i in range(0, len(myStr)):
    if(myStr[i]==" "or my_str[i]=="."):
        output = myStr[start:i] + " "+ output
        start = i+1
print(output)
```

```
peppers pickled of peck a picked piper peter
```

In [27]:

```python
#Write a python program to find below output using loop:-
```

```python
#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:- '.sreppep delkcip fo kcep a dekcip repip retep'

str = 'peter piper picked a peck of pickled peppers.'
def reverse_string(str):
    str1 = ""   # Declaring empty string to store the reversed string
    for i in str:
        str1 = i + str1
    return str1    # It will return the reverse string to the caller function

print("The original string is: ",str)
print("The reverse string is:",reverse_string(str))
```

```
The original string is:  peter piper picked a peck of pickled peppers.
The reverse string is: .sreppep delkcip fo kcep a dekcip repip retep
```

In [32]:

```python
#Write a python program to find below output using loop:-

#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:- 'retep repip dekcip a kcep fo delkcip sreppep'

my_str = 'peter piper picked a peck of pickled peppers'

def reverseWordSentence(my_str):

    # Splitting the Sentence into list of words.
    words = my_str.split(" ")

    # Reversing each word and creating
    # a new list of words
    # List Comprehension Technique
    newWords = [word[::-1] for word in words]

    # Joining the new list of words
    # to for a new Sentence
    newSentence = " ".join(newWords)

    return newSentence

print(reverseWordSentence(my_str))




##  2nd method
myStr = 'peter piper picked a peck of pickled peppers.'
output = ''
reverseWord =''
start=0
for i in range(0,len(myStr)):
    if(myStr[i]==" " or myStr[i]=="."):
        word = myStr[start:i]
        reverseWord = word[-1:-len(word)-1:-1] + " "
        output +=reverseWord
        start = i+1
```

```
print(output)
```

```
retep repip dekcip a kcep fo delkcip sreppep
```

In [37]:

```
#Write a python program to find below output using loop:-

#Input:- 'peter piper picked a peck of pickled peppers.'
#Output:- 'Peter Piper Picked A Peck Of Pickled Peppers'

#using code
my_str = 'peter piper picked a peck of pickled peppers.'
output = my_str.title()
print(output)

#Using if else and def function using strip and capitalize function

def capitalize_each_word(original_str):
    result = ""
    # Split the string and get all words in a list
    list_of_words = original_str.split()
    # Iterate over all elements in list
    for elem in list_of_words:
        # capitalize first letter of each word and add to a string
        if len(result) > 0:
            result = result + " " + elem.strip().capitalize()
        else:
            result = elem.capitalize()
    # If result is still empty then return original string else returned capitalize
    if not result:
        return original_str
    else:
        return result
sample_text = 'peter piper picked a peck of pickled peppers.'
result = capitalize_each_word(sample_text)
print(result)



#2nd method

myStr = 'peter piper picked a peck of pickled peppers.'
output =''
start = 1
for i in range(0, len(myStr)):
    if(" " == myStr[i]):
        output = output + myStr[start-1].upper() +myStr[start:i] + " "
        start = i+2

output = output + myStr[start-1].upper() +myStr[start:i]
print(output)
```

```
Peter Piper Picked A Peck Of Pickled Peppers.
Peter Piper Picked A Peck Of Pickled Peppers.
```

In [16]:

```
#Write a python program to find below output using loop:-
```

```
#Input:- 'Peter Piper Picked A Peck Of Pickled Peppers.'
#Output:- 'Peter piper picked a peck of pickled peppers'

# Using code .capitalize()

my_str = 'Peter Piper Picked A Peck Of Pickled Peppers.'
print(my_str.capitalize())


#using loop

myStr = 'peter piper picked a peck of pickled peppers.'
output =''
start = 0
for i in range(0, len(myStr)):
    if(" " == myStr[i]):
        output = output + myStr[start:i] + " "
        start = i+1
output = output + myStr[start:i]
output = output[0].upper() +output[1:]
print(output)
```

```
Peter piper picked a peck of pickled peppers.
Peter piper picked a peck of pickled peppers
```

In [17]:

```
#Write a python program to implement index method using loop. If sub_str is found i
# of first occurrence of first character of matching string in my_str:-

#Input:- my_str = 'Peter Piper Picked A Peck Of Pickled Peppers.', sub_str = 'Pickl
#Output:- 29

my_str ='Peter Piper Picked A Peck Of Pickled Peppers.'
sub_str ="Pickl"

for i in range(0,len(my_str)):
    if my_str[i:i+len(sub_str)] == sub_str:
        print(i)
```

```
29
```

In [54]:

```
#Write a python program to implement replace method using loop. If sub_str is found
#occurrence of sub_str with new_str else it will will print sub_str not found:-

#Input:- my_str = 'Peter Piper Picked A Peck Of Pickled Peppers.', sub_str = 'Peck'
#Output:- 'Peter Piper Picked A Pack Of Pickled Peppers.'

my_str = 'Peter Piper Picked A Peck Of Pickled Peppers.'
sub_str = 'Peck'
new_str1 = 'Pack'
out_str = ''
j = 0
new_str = ''
for i in range(len(my_str)):
    if my_str[i] == sub_str[j]:
        new_str += my_str[i]
        j += 1
        if len(new_str) == len(sub_str) and new_str == sub_str:
```

```
            ind = i-(len(sub_str)-1)
            print(ind)
            out_str = my_str[:ind] + new_str1 + my_str[ind+(len(sub_str)):]
            print(out_str)
            break
        elif len(new_str) == len(sub_str) and new_str != sub_str:
            new_str = ''
    elif i == (len(my_str)-1):
        print('string not found!')
    else:
        new_str = ''
        j = 0


# 2nd Method

a = 'Peter Piper Picked A Peck Of Pickled Peppers.'
list1 = list(a)
b = 'Peck'
newstr = 'pack'
c =0
if b in a:
        d = a.index(b)
for i in range(d,d+len(b)):
    list1[i]=newstr[c]
    c+=1
''.join(list1)
```

```
21
Peter Piper Picked A Pack Of Pickled Peppers.
```

In [62]:

```
#Write a python program to find below output (implements rjust and ljust) using loo

#Input:- 'Peter Piper Picked A Peck Of Pickled Peppers.', sub_str = 'Peck',
#Output:- '********************Peck********************'

my_str='Peter Piper Picked A Peck Of Pickled Peppers.'
sub_str='Peck'
out_str=''
i=0
while i in range(0,len(my_str),1):
    if my_str[i:(i+len(sub_str)):1]==sub_str:
        out_str=out_str+sub_str
        i = i+(len(sub_str)-1)
    else:
        out_str=out_str+'*'
    i=i+1
print(out_str)
```

```
********************Peck********************
```

In [19]:

```
#Write a python program to find below output using loop:-

#Input:- 'This is Python class', sep = 'is',
#Output:- ['This', 'is', 'Python class']
```

```python
inp_str='This is Python class'
sep='is'
out_str=''
l_str=''
r_str=''
length=0
out_lst=[]
for letter in inp_str:
    length+=1
    out_str += letter
    if sep in out_str:
        l_str=inp_str[:length-len(sep):]
        r_str=inp_str[length::]
        out_str=''
out_lst.append(l_str)
out_lst.append(sep)
out_lst.append(r_str)
print(out_lst)

# my_str='This is Python Class '
# my_str1 =''
# my_str2 =''
# my_lst =[]
# sep = 'is'
# x=0
# for i in my_str:

#     if i !=' ':
#         my_str1 += i
#     else:
#         if sep not in my_lst:
#             my_lst += [my_str1]
#             my_str1 = ''
#         else:
#             my_str1=my_str1 + ' '
# my_lst += [my_str1.strip()]
# print(my_lst)




# my_str='This is Python class'
# sep_str=' is'
# out_str=''
# out_lst=[]
# i=0
# x=True
# # print(my_str.partition(' is '))
# while i in range(0,(len(my_str)),1):
#     if my_str[i]==' ' and x==True and i<=(len(my_str)):
#         out_lst.append(out_str)
#         out_str=''
#     if my_str[i:(i+len(sep_str)):1]==sep_str:
#             out_lst.append(sep_str)
#             out_str=''
#             i = i+(len(sep_str))
```

```
#             x=False
#     if i ==(len(my_str)):
#         out_lst.append(out_str)
#     else:
#         out_str=out_str+my_str[i]
#     i=i+1
# out_lst.append(out_str)
# print(out_lst)


# Alternative method

sentence = 'This is Python class'
sep = ' is'
tmp = ''
out = ''
out_lst = []
for i in sentence:
    if i == ' ':
        if tmp == sep:
            out_lst.append(out)
            out_lst.append(tmp)
            out = ''
        else:
            out += tmp
        tmp = ' '
    else:
        tmp += i

if tmp:
    out += tmp
    out_lst.append(out[1:])

print(out_lst)


# Another method

str1='This is Python class'
sep='is'
word=''
out=[]
for i in (str1):
    if i!=' ':
        word+=i
    elif sep not in out:
        out+=[word]
        word=''
    elif sep  in out:
        word+=i

else:
    out+=[word]
print(out)
```

```python
# Another method
for i in range(len(my_str)):
    if my_str[i:i+len(sep)] == sep:
        output.append(my_str[start:i])
        output.append(sep.strip())
        start = i + len(sep)
output.append(my_str[start:])
print(output)
```

```
['This ', 'is', ' Python class']
```